

Now,
$$I_2' = \frac{V_1}{\sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}} = \frac{400/\sqrt{3}}{\sqrt{(0.4 + 0.6)^2 + (1 + 1)^2}} = 103.3 \text{ A}$$

$\therefore T_{g-max} = \frac{3 \times 103.3^2 \times 1/0.29}{2\pi \times 1500/60} = 351 \text{ N-m} \quad \dots \text{ assuming } N_s = 1500 \text{ r.p.m.}$

(iii) The equivalent circuit for one phase for a slip of 0.05 is shown in Fig. 32.59 (b).

$$I_2' = 231 / [(20 + 0.4) + j 2] = 11.2 - j 1.1$$

$$I_0 = 231 / (10 + j 50) = 0.89 - j 4.4$$

$$I_1 = I_0 + I_2' = 12.09 - j 5.5 = 13.28 \angle -24.4^\circ; \text{ p.f.} = \cos 24.4^\circ = 0.91 \text{ (lag)}$$

Tutorial Problem No. 34.4.

1. A 3-phase, 115-volt induction motor has the following constants : $R_2 = 0.07 \Omega$; $R_2' = 0.08 \Omega$, $X_1 = 0.4 \Omega$ and $X_2' = 0.2 \Omega$. All the values are for one phase only. At which slip the gross power output will be maximum and the value of the gross power output ? [11.4% ; 8.6 kW]
2. A 3-phase, 400-V, Y-connected induction motor has an equivalent T-circuit consisting of $R_1 = 1 \Omega$, $X_1 = 2 \Omega$, equivalent rotor values are $R_2' = 1.2 \Omega$, $X_2' = 1.5 \Omega$. The exciting branch has an impedance of $(4 + j 40) \Omega$. If slip is 5% find (i) current (ii) efficiency (iii) power factor (iv) output. Assume friction loss to be 250 W. [(i) 10.8 A (ii) 81% (iii) 0.82 (iv) 5 kW]
3. A 50 HP, 440 Volt, 3-phase, 50 Hz Induction motor with star-connected stator winding gave the following test results:
 (i) No load test: Applied line voltage 440 V, line current 24 A, wattmeter reading 5150 and 3350 watts.
 (ii) Blocked rotor test: applied line voltage 33.6 volt, line current 65 A, wattmeter reading 2150 and 766 watts.
 Calculate the parameters of the equivalent circuit.

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[(i) Shunt branch : $R_o = 107.6 \text{ ohms}$, $X_m = 10.60 \text{ ohms}$ (ii) Series branch : $r = 0.23 \text{ ohm}$, $x = 0.19 \text{ ohm}$]

OBJECTIVE TESTS – 34

1. Regarding skewing of motor bars in a squirrel-cage induction motor, (SCIM) which statement is false ?
 (a) it prevents cogging
 (b) it increases starting torque
 (c) it produces more uniform torque
 (d) it reduces motor 'hum' during its operation.
2. The principle of operation of a 3-phase. Induction motor is most similar to that of a
 (a) synchronous motor
 (b) repulsion-start induction motor
 (c) transformer with a shorted secondary
 (d) capacitor-start, induction-run motor.
3. The magnetising current drawn by transformers and induction motors is the cause of theirpower factor.
 (a) zero (b) unity
 (c) lagging (d) leading.
4. The effect of increasing the length of air-gap in an induction motor will be to increase the
 (a) power factor
 (b) speed
 (c) magnetising current
 (d) air-gap flux.
(Power App-II, Delhi Univ. Jan. 1987)
5. In a 3-phase induction motor, the relative speed of stator flux with respect tois zero.
 (a) stator winding (b) rotor
 (c) rotor flux (d) space.
6. An eight-pole wound rotor induction motor operating on 60 Hz supply is driven at 1800 r.p.m. by a prime mover in the opposite direction of revolving magnetic field. The frequency of rotor current is
 (a) 60 Hz (b) 120 Hz
 (c) 180 Hz (d) none of the above.

(Elect. Machines, A.M.I.E. Sec. B, 1993)

7. A 3-phase, 4-pole, 50-Hz induction motor runs at a speed of 1440 r.p.m. The rotating field produced by the rotor rotates at a speed ofr.p.m. with respect to the rotor.
(a) 1500 (b) 1440
(c) 60 (d) 0.
8. In a 3- ϕ induction motor, the rotor field rotates at synchronous speed with respect to
(a) stator (b) rotor
(c) stator flux (d) none of the above.
9. Irrespective of the supply frequency, the torque developed by a SCIM is the same whenever is the same.
(a) supply voltage (b) external load
(c) rotor resistance (d) slip speed.
10. In the case of a 3- ϕ induction motor having $N_s = 1500$ rpm and running with $s = 0.04$
(a) revolving speed of the stator flux is space isrpm
(b) rotor speed isrpm
(c) speed of rotor flux relative to the rotor isrpm
(d) speed of the rotor flux with respect to the stator isrpm.
11. The number of stator poles produced in the rotating magnetic field of a 3- ϕ induction motor having 3 slots per pole per phase is
(a) 3 (b) 6
(c) 2 (d) 12
12. The power factor of a squirrel-cage induction motor is
(a) low at light loads only
(b) low at heavy loads only
(c) low at light and heavy loads both
(d) low at rated load only.
- (Elect. Machines, A.M.I.E. Sec. B, 1993)
13. Which of the following rotor quantity in a SCIM does NOT depend on its slip?
(a) reactance (b) speed
(c) induced emf (d) frequency.
14. A 6-pole, 50-Hz, 3- ϕ induction motor is running at 950 rpm and has rotor Cu loss of 5 kW. Its rotor input iskW.
(a) 100 (b) 10
(c) 95 (d) 5.3.
15. The efficiency of a 3-phase induction motor is approximately proportional to
(a) $(1-s)$ (b) s
(c) N (d) N_s .
16. A 6-pole, 50-Hz, 3- ϕ induction motor has a full-load speed of 950 rpm. At half-load, its speed would berpm.
(a) 475 (b) 500
(c) 975 (d) 1000
17. If rotor input of a SCIM running with a slip of 10% is 100 kW, gross power developed by its rotor is kW.
(a) 10 (b) 90
(c) 99 (d) 80
18. Pull-out torque of a SCIM occurs at that value of slip where rotor power factor equals
(a) unity (b) 0.707
(c) 0.866 (d) 0.5
19. Fill in the blanks.
When load is placed on a 3-phase induction motor, its
(i) speed
(ii) slip
(iii) rotor induced emf
(iv) rotor current
(v) rotor torque
(vi) rotor continues to rotate at that value of slip at which developed torque equals torque.
20. When applied rated voltage per phase is reduced by one-half, the starting torque of a SCIM becomes of the starting torque with full voltage.
(a) $1/2$ (b) $1/4$
(c) $1/\sqrt{2}$ (d) $\sqrt{3}/2$
21. If maximum torque of an induction motor is 200 kg-m at a slip of 12%, the torque at 6% slip would be kg-m.
(a) 100 (b) 160
(c) 50 (d) 40
22. The fractional slip of an induction motor is the ratio
(a) rotor Cu loss/rotor input
(b) stator Cu loss/stator input
(c) rotor Cu loss/rotor output
(d) rotor Cu loss/stator Cu loss
23. The torque developed by a 3-phase induction motor depends on the following three factors:
(a) speed, frequency, number of poles
(b) voltage, current and stator impedance
(c) synchronous speed, rotor speed and frequency

- (d) rotor emf, rotor current and rotor p.f.
24. If the stator voltage and frequency of an induction motor are reduced proportionately, its
- (a) locked rotor current is reduced
 - (b) torque developed is increased
 - (c) magnetising current is decreased
 - (d) both (a) and (b)
25. The efficiency and p.f. of a SCIM increases in proportion to its
- (a) speed
 - (b) mechanical load
 - (c) voltage
 - (d) rotor torque
26. A SCIM runs at **constant** speed only so long as
- (a) torque developed by it remains constant
 - (b) its supply voltage remains constant
 - (c) its torque exactly equals the mechanical load
 - (d) stator flux remains constant
27. The synchronous speed of a linear induction motor does NOT depend on
- (a) width of pole pitch
 - (b) number of poles
 - (c) supply frequency
 - (d) any of the above
28. Thrust developed by a linear induction motor depends on
- (a) synchronous speed
 - (b) rotor input
 - (c) number of poles
 - (d) both (a) and (b)

ANSWERS

1. b 2. c 3. c 4. c 5. c 6. c 7. c 8. a 9. d 10. (i) 1500 (ii) 1440 (iii) 60 (iv) 1500 11. b 12. a
13. b 14. a 15. a 16. c 17. b 18. b 19. (i) decreases (ii) increases (iii) increases (iv) increases
(v) increases (vi) applied 20. b 21. b 22. a 23. d 24. d 25. b 26. c 27. b 28. d